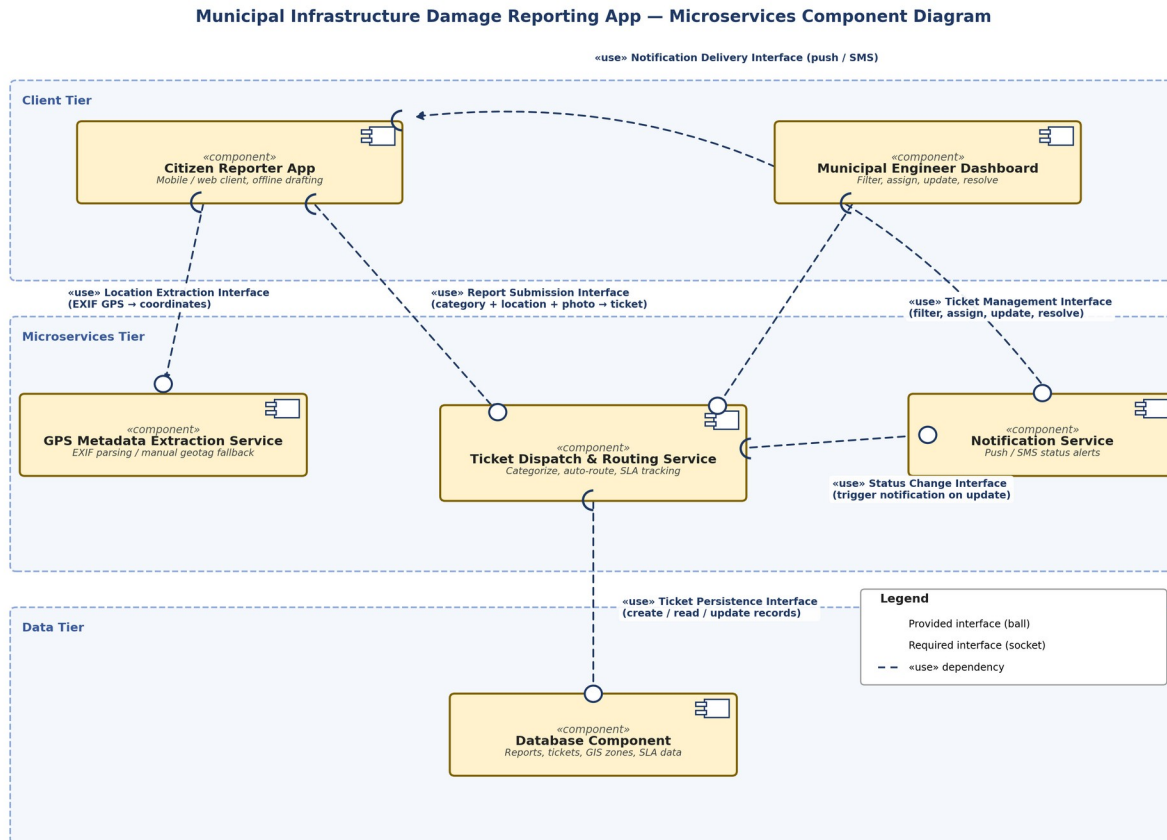


Lab 3: Component Modelling & Architectural Pattern Selection

Municipal Infrastructure Damage Reporting App

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1. UML Component Diagram



2. Written Justification

Architecture Selection

We chose the Microservices Architecture for the Municipal Infrastructure Damage Reporting App.

Reasons for This Choice

1. Independent scalability for city-wide peak load: NFR-002 requires the system to scale to at least 50,000 concurrent citizen users while keeping 95th-percentile response times under 3 seconds, even on low-end devices over 2G/3G networks. A Microservices Architecture lets the high-traffic, citizen-facing services — GPS Metadata Extraction and Ticket Dispatch & Routing — scale out independently during reporting spikes (e.g., after a storm damages multiple roads at once), without over-provisioning the lower-traffic Municipal Engineer Dashboard or Notification Service.
2. Fault isolation across independently-evolving workflows: the system serves two distinct actor workflows — Citizen Reporter submission/tracking and Municipal Engineer ticket management — that change and deploy on different schedules. Splitting these into separate services means an outage or slow deployment in the Notification Service does not block citizens from submitting new damage reports (FR-001–FR-003), and new SLA/escalation logic for engineers can be released to the Ticket Dispatch & Routing Service without redeploying the citizen-facing app.

Security Advantage

Confining GPS/photo metadata handling to its own GPS Metadata Extraction Service, reached only through a defined interface, and scoping each service's database access independently, limits the blast radius of a compromise in the lower-trust, publicly-accessible Citizen Reporter App. A breach there cannot directly reach the Database Component or the Municipal Engineer Dashboard. This service-boundary isolation also gives each microservice its own point to enforce TLS/authentication on data in transit, directly supporting NFR-001's requirement that synced data be secured against tampering.

Performance Benefit

Because the Ticket Dispatch & Routing Service and Notification Service can be scaled and deployed independently of the Citizen App and Engineer Dashboard, sudden spikes in report submissions (e.g., after severe weather) can be absorbed by horizontally scaling just those two services. This lets the system sustain 50,000 concurrent users within the required 3-second response threshold (NFR-002) without over-provisioning the entire stack, as a monolithic Client-Server design would require.